

THE ARMORY

SHOOTING SPORTS CLUB

BUILD SPECIFICATION

The Armory Management System — implementation specification for the development team

Prepared for **Development team**

Client **The Armory Shooting Sports Club — Abuja National Stadium, FCT**

Document type **Implementation specification. Buildable, estimable, testable**

Rationale document **Design Specification Rev. 2 — read once, referenced throughout**

Phase 1 effort **Approximately 225–315 person-days**

Status **Ready for estimation. Blocking items listed in Section 14**

SECTION 1

How to read this document

This is the build document. It states what to construct, in what order, and what "finished" means for each part. It does not restate the reasoning behind the product decisions.

The companion Design Specification carries that reasoning and should be read once before starting. Where this document says a rule must be enforced, that document explains what happens to the business if it is not. Developers who understand the second document make better decisions in the gaps the first one inevitably leaves.

1.1 The business, in six lines

- A private shooting sports club in Abuja. Closed membership. No public access of any kind.
- A person may be on the premises only as a member, or as the named guest of a member whose tier permits hosting, and only while that host is present.
- Membership is by application and admission, not purchase. The roster has a hard cap; a waitlist sits behind it.
- Some members may bring firearms they own. Every firearm on site, whoever owns it, is in one register with one movement log.
- The club is run by the founder and a handful of range officers. There is no administrator and there will not be one.
- Power and internet are unreliable. The club must be able to operate through an outage.

1.2 The five rules that override convenience

When a design choice is unclear, these decide it

1. Members and guests enter their own data, before arrival. Range officers confirm and issue. Budget: fifteen seconds per shooter at the desk.
2. The desk and lane surfaces work fully offline. Not read-only — fully.
3. Custody events, waiver signatures and scores are append-only. Nothing is edited; nothing is deleted; a correction is a new row referencing the old one.
4. Tier is a permission matrix evaluated by code, never a label rendered on a page.
5. Money from a guest visit lands on the host. A guest never transacts with the club.

SECTION 2

Stack and environment

Specified where the choice is load-bearing; left to the team where it is not. Substitutions are acceptable provided the stated property is preserved.

Layer	Specification	Non-negotiable property
Client	Progressive web application. Single codebase, four surfaces, installable to home screen. Framework at the team's discretion.	No native builds. No app store. Updates reach club tablets on next open.
Local store	IndexedDB, via a wrapper of the team's choosing.	Survives tab close, app close, device restart and power loss.
Service worker	Full application shell and asset caching; background sync where supported.	Cold start with no network must reach a usable desk screen in under one second.
Server	Any mature server runtime. Team's choice.	Idempotent write endpoints. See Section 8.
Database	PostgreSQL.	Relational integrity, transactional writes, JSONB for shot detail and audit payloads.
File storage	Object storage, separate from the application database.	Licence scans and photographs are not in the primary database. Signed, expiring URLs only.
Hosting	Region with low latency to Nigeria. Managed platform preferred over self-managed.	Automated daily off-site backup with a tested restore.
Money	All amounts stored as integer kobo. Never floats.	NGN only at launch.
Time	All timestamps stored UTC. Rendered Africa/Lagos.	Client-generated timestamps are recorded alongside server receipt time, never instead of it.
IDs	UUIDv7 for all primary keys, generated client-side where the record can originate offline.	Offline-created records must not require a server round trip to obtain an identity.

2.1 Environments

Three: local, staging, production. Staging must hold realistic seed data — a hundred members, a full tier matrix, several hundred sessions, a populated firearm register — because most defects in this system only appear at volume or at a period boundary.

SECTION 3

Data model

Field lists are the minimum. Add what the implementation needs; do not remove. Every table carries created_at, updated_at, and where noted, an append-only constraint.

3.1 People and standing

Table	Fields	Constraints
people	id, first_name, last_name, phone, email, date_of_birth, photo_url, address, emergency_contact_name, emergency_contact_phone, notes	phone unique, E.164. One row per human — applicant, guest and member are states, not separate tables.
memberships	id, person_id, tier_id, member_number, status, is_founding, principal_membership_id, started_on, renews_on, ended_on	member_number unique, sequential, never reused. status: pending, active, lapsed, suspended, resigned. principal_membership_id set only for attached (Partner) memberships.
tiers	id, name, monthly_fee_kobo, annual_fee_kobo, can_host, guest_allowance_annual, guest_concurrent_max, can_own_firearm, can_store_firearm, discipline_access, booking_horizon_days, concurrent_bookings_max, is_attachable, sort_order, active	Explicit columns, not a JSON blob. These are read by enforcement code and must be queryable and constrainable.
applications	id, person_id, sponsor_type, sponsor_membership_id, requested_tier_id, status, submitted_at, decided_by_staff_id, decided_at, decision_note	sponsor_type: member, club. status: submitted, under_review, admitted, waitlisted, declined, withdrawn.
waitlist_entries	id, application_id, position, added_at, last_contacted_at, status	position maintained server-side. Never derived from a client sort.
waiver_versions	id, version, body, effective_from, is_active	Exactly one active version at a time.
waiver_signatures	id, person_id, waiver_version_id, signature_image_url, signed_at, device_id, recorded_at	APPEND-ONLY. No update, no delete. A resigature is a new row.
qualifications	id, person_id, discipline, level, assessed_by_staff_id, assessed_at, expires_at	Club competency. Distinct from licences.

Table	Fields	Constraints
licences	id, person_id, licence_number, issuing_authority, calibres, issued_on, expires_on, document_url, status, verified_by_staff_id, verified_at	status: pending_review, verified, expired, revoked. Capability activates only on verified.
staff_users	id, person_id, role, pin_hash, certifications, active	role: founder, range_officer, read_only. No shared accounts — see 11.2.
devices	id, label, surface, registered_by_staff_id, registered_at, last_seen_at, revoked_at	surface: desk, lane. Desk and lane surfaces load only on a registered, unrevoked device.

3.2 Hosting

This cluster is the acquisition funnel. Treat its correctness as equal in importance to the firearm register.

Table	Fields	Constraints
host_allowances	id, membership_id, period_start, period_end, included_quota, used_count, overage_count	One row per membership per period. used_count is authoritative and mutated only server-side, inside a transaction. See 8.3.
guest_invitations	id, host_membership_id, guest_person_id, booking_id, token, status, issued_at, opened_at, completed_at, cancelled_at, is_chargeable	token unique, single-use, expiring. status: issued, opened, completed, attended, cancelled, expired. Cancellation before the session returns the allowance.
guest_visit_summary	person_id, visit_count, first_visit_at, last_visit_at, distinct_host_count, converted_membership_id	Materialised view or maintained table. Counts across all hosts, not per host. Drives the conversion prompt.

3.3 Activity

Table	Fields	Constraints
lanes	id, discipline, number, status, position_capacity	status: available, maintenance, closed.
bookings	id, host_membership_id, booking_date, slot_start, slot_end, discipline, status, created_at, cancelled_at	status: draft, confirmed, cancelled, completed, no_show. Only a member may hold a booking.
booking_participants	id, booking_id, person_id, role, guest_invitation_id	role: member_shooter, guest_shooter, spectator. Every guest_shooter must carry a guest_invitation_id.
sessions	id, booking_id, session_date, opened_at, opened_by_staff_id, closed_at, closed_by_staff_id, status	A session cannot close while anything issued against it is outstanding. See 6.4.
participations	id, session_id, person_id, role, host_person_id, lane_id, tier_snapshot, checked_in_at, checked_in_by_staff_id, checked_out_at	APPEND-ONLY on check-in. host_person_id required when role is guest_shooter. tier_snapshot records the tier as at the visit — never join to the live tier for historical reporting.
rounds	id, participation_id, discipline, format, total_score, shot_detail, capture_method, source_reference, captured_by_staff_id, captured_at	APPEND-ONLY. capture_method: manual, electronic. shot_detail JSONB, nullable.

Table	Fields	Constraints
incidents	id, session_id, category, narrative, occurred_at, reported_by_staff_id, follow_up_status	Plus incident_persons join carrying person_id and involvement. Must be creatable offline.

3.4 Property

ARCHITECTURAL
INSTRUCTION

One register.
One log.
No exceptions.

Club-owned and member-owned firearms are rows in the same table, distinguished by an ownership column. Do not create a second table, a second module, or a parallel workflow.

custody_events is append-only and is the sole source of truth for where any firearm is. Firearm status is derived from it, not maintained independently — if the two can disagree, they eventually will.

Table	Fields	Constraints
-------	--------	-------------

firearms	id, serial_number, make, model, calibre, discipline, ownership, owner_person_id, licence_id, status, condition, round_count, acquired_on, decommissioned_on	serial_number unique. ownership: club, member. owner_person_id and licence_id required when ownership is member. status derived from custody_events.
custody_events	id, firearm_id, event_type, actor_staff_id, counterparty_person_id, session_id, occurred_at, recorded_at, device_id, corrects_event_id, note	APPEND-ONLY. event_type: issued, returned, brought_in, taken_out, stored, withdrawn, sent_for_service, returned_from_service, decommissioned, correction. No update. No delete. Ever.
ammunition_lots	id, calibre, quantity_received, quantity_remaining, supplier, received_on, reference	quantity_remaining derived from issues, not decremented independently.
ammunition_issues	id, lot_id, participation_id, qty_issued, qty_fired, qty_returned, issued_by_staff_id, reconciled_at	Reconciled per participation, not per day. qty_issued must equal qty_fired plus qty_returned at reconciliation.
storage_assignments	id, firearm_id, membership_id, cabinet_reference, assigned_on, released_on, fee_status	Build the table; leave the workflow behind a feature flag until the storage decision lands. See 14.
assets	id, category, identifier, status, condition	Non-firearm equipment. Issue and return tracked without custody-log rigour.

3.5 Money

Table	Fields	Constraints
charges	id, payer_person_id, reference_type, reference_id, line_items, total_kobo, is_settled, settled_payment_id, created_at	payer_person_id for a guest charge is ALWAYS the host. Enforce in code, not convention.
payments	id, person_id, amount_kobo, purpose, method, gateway, gateway_reference, status, paid_at	purpose: subscription, guest_overnight, account_topup, bar, retail, other. gateway_reference unique where present.
accounts	id, person_id, balance_kobo	One per person. Balance derived from account_transactions, never written directly.
account_transactions	id, account_id, amount_kobo, direction, reference_type, reference_id, created_at	APPEND-ONLY. Ledger, not a mutable balance.

3.6 Audit

Table	Fields	Constraints
audit_log	id, actor_staff_id, action, entity_type, entity_id, before, after, device_id, override_reason, occurred_at, recorded_at	APPEND-ONLY. Every custody event, score, waiver, admission, guest authorisation, block and override. An override is permitted but never silent — override_reason is mandatory when present.

SECTION 4

The capability model

One service, called from every surface. No screen makes its own permission decision, and no permission check is duplicated in the client and the server — the client calls it to decide what to show, the server calls it to decide what to allow.

4.1 The interface

evaluate(person, capability, context) → { allowed, reason, remedy }

allowed — boolean.

reason — a stable machine code plus a human string. The human string is what the desk displays and what the member reads in the portal, so it must be written by someone who can write.

remedy — optional. What resolves it. "Renew in the portal." "Your host has not arrived." "Licence expired 14 March."

4.2 Capabilities to implement

Capability	Evaluated at	Fails when
MAY_BOOK	Booking	Membership not active; booking horizon exceeded; concurrent booking limit reached; discipline not in tier access.
MAY_HOST	Booking, and again at check-in	Tier cannot host; allowance exhausted and overage declined; concurrent guest limit exceeded for the session.
MAY_ATTEND	Check-in	Membership inactive; waiver missing, superseded or expired; suspension open.
GUEST_MAY_ATTEND	Check-in	No valid invitation; invitation not completed; waiver not signed; HOST NOT YET CHECKED IN.
MAY_SHOOT_DISCIPLINE	Check-in and lane	Discipline outside tier access; required qualification missing or expired.
MAY_USE_OWN_FIREARM	Check-in and lane	Tier lacks capability; no verified licence; licence expired; firearm not registered to this person; calibre outside licence.
MAY_STORE_FIREARM	Storage assignment	Tier lacks capability; feature disabled pending the storage decision.

The host-presence rule in GUEST_MAY_ATTEND is easy to miss and is not optional. A guest who arrives before their host waits. The check must re-evaluate when the host checks in, so the desk row clears by itself rather than requiring the officer to retry.

4.3 Blocks are a product surface

- Every block returns a specific single-line reason. Never "not permitted", never a code, never a stack of conditions.
- Blocks that a member could have resolved in advance must have been surfaced in the portal in advance. Anything a member first learns at the desk, in front of their guest, is a defect in this system.
- An officer may override a block where the override is permitted. The override records who, why, and on which device, and appears on the founder's dashboard the same day.

SECTION 5

State machines

Implement these as explicit transitions with guards. Do not allow status to be set by direct assignment from a form.

Entity	Transitions
Application	submitted → under_review → admitted waitlisted declined. waitlisted → admitted withdrawn. Any → withdrawn. Admission requires roster headroom or an explicit founder override.
Membership	pending → active (on payment) → lapsed (renewal missed) suspended (staff action) resigned. lapsed → active on payment. suspended → active on staff action. A Partner membership cannot outlive its principal.
Booking	draft → confirmed → completed cancelled no_show. Cancellation releases allowance and lane capacity. no_show set by the end-of-day close, not manually.
GuestInvitation	issued → opened → completed → attended. issued opened completed → cancelled (returns allowance) expired. attended is terminal and does not return allowance.
Session	open → closed. Close is GUARDED: blocked while any firearm issued against the session is unreturned, any ammunition issue unreconciled, or any participation without checkout.
Firearm	Status is DERIVED from the latest custody_event. Never set directly. in_service, issued, off_site, in_storage, at_service, decommissioned.
Licence	pending_review → verified → expired (by date) revoked (staff action). Capability is live only while verified and unexpired.

SECTION 6

Screens and acceptance criteria

Screen inventory by surface. Acceptance criteria are the conditions for "done" — they are written to be turned into tests.

6.1 Public site

The smallest part of the build. Static or near-static: positioning, the building, disciplines, leagues, how admission works, register interest. One live form, writing an application at sponsor_type = club.

6.2 Member portal

Screen	Acceptance
Home	Next booking, remaining guest allowance, most recent score, anything expiring within 60 days. Loads in under two seconds on a mid-range Android over 3G.
Book a session	Availability computed from lanes, hours, officer coverage and existing bookings — never a manually maintained calendar. Naming a guest calls MAY_HOST before the booking can be confirmed. Allowance shown before and after.
Invite a guest	Creates the invitation, sends the link over WhatsApp, decrements allowance inside the same transaction as the booking write. Cancelling returns it.
My documents	Waiver status and version; licence upload with expiry and verification state; owned firearm registrations.
My shooting	Score history, personal best by discipline, trend. Phase 1 shows history; Phase 2 adds standing.
Account	Charges, payments, balance, renewal date, tier and its privileges stated plainly.
Sponsor an application	A member puts a former guest forward. Pre-fills from the person record and their visit history.

6.3 Guest link

HIGHEST-STAKES SCREEN IN THE BUILD

Two minutes, on a phone, in poor signal, by someone who has never heard of you

This is the first surface a future member ever touches. It is opened once, in a car, by someone slightly nervous about shooting for the first time.

Acceptance: personal details, emergency contact, date of birth and a signed waiver, completed end to end in under two minutes on a mid-range Android over 3G, with no account creation and no password.

Also carries: what to expect, what to wear, that nothing needs to be brought, that no experience is required.

After the session it becomes their result page — their score, and a route to apply.

6.4 Desk application

Screen	Acceptance
Today	The default and dominant screen. Expected arrivals in time order: name, photograph, tier or host, discipline, and ONE status indicator — clear, attention, blocked. Renders in under one second from cold, with no network. Status distinguishable by shape and position as well as colour.
Person detail	Opened only when the officer needs it. Documents, history, current blocks with reasons.
Check in	Three taps maximum from Today to checked in with a lane assigned. Calls the capability service; a block states its reason in one line. Guest rows re-evaluate automatically when the host checks in.
Issue equipment	Firearm by serial and ammunition by quantity, written as custody and ammunition rows against the participation. Works offline.
Session close	Guarded. Lists everything outstanding. Cannot be forced without an override carrying a reason.
End of day	One screen reconciling all sessions: firearms out versus in, ammunition issued versus fired versus returned, participations without checkout, sync queue depth. The range should not be locked until it clears.

Explicitly not present: walk-in enrolment. There is no path at the desk to create a person who was not authorised in advance. If an officer needs one, the answer is a booking, not a form.

6.5 Lane application

Screen	Acceptance
Relay	Who is on which lane right now. Large type, readable at arm's length in daylight.
Pair	Assign shooter to firearm by serial. Calls MAY_USE_OWN_FIREARM. Completable in under twenty seconds, standing, one-handed.
Score	Select shooter, enter total, confirm. No free-text entry in the normal path. Under twenty seconds. Guests are scored identically to members.
Incident	Always reachable, never more than one tap away, always available offline.

6.6 Owner dashboard

Roster against cap with waitlist depth; hosting rate; guest-to-application conversion; admissions by host; occupancy by day and slot separating member and guest load; lapse list; revenue by source; reconciliation exceptions; expiries within 90 days; overrides in the last 7 days.

API surface

Indicative resources, not a contract. Shape it as the team prefers. Two properties are mandatory.

Mandatory properties of every write endpoint

IDEMPOTENT. Every write accepts a client-generated UUIDv7 as the record identity and is safe to replay. A queued write delivered twice must produce one row, not two. This is the single most important API requirement in the system.

ATTRIBUTE. Every write records actor, device and both client and server timestamps.

Group	Resources
Identity	POST /auth/otp/request, /auth/otp/verify, /auth/staff/unlock; GET /me

Admission	applications (create, list, decide); waitlist (list, reorder, notify)
Members	people; memberships; tiers; qualifications; licences (upload, verify)
Hosting	allowances (read); guest-invitations (create, cancel); /visit/{token} (public read and complete)
Booking	availability (query); bookings; booking-participants
Operations	sessions (open, close); participations (check-in, check-out); lanes
Armoury	firearms; custody-events (create only); ammunition-lots; ammunition-issues
Scoring	rounds (create only); score-import (adapter endpoint, Phase 2)
Money	charges; payments; paystack webhook; accounts; account-transactions
Sync	GET /sync/daypack; POST /sync/push; GET /sync/pull
Reporting	dashboard metrics; exceptions; expiries

SECTION 8

Offline and synchronisation

The largest single line of effort in the build and the one that produces no visible feature. Budget 25–35 person-days and do not let it be compressed — an offline layer that mostly works is worse than none, because staff will trust it.

8.1 The day pack

On connection, each desk and lane device pulls and caches: today's and tomorrow's expected arrivals with computed status; all active memberships and tiers; all guest invitations for the window; the full firearm register; ammunition lot balances; the active waiver version; staff and device records. Refreshed continuously while online; sufficient to run a full day alone.

8.2 Two classes of data

Class	Tables	Strategy
Append-only	custody_events, waiver_signatures, rounds, ammunition_issues, participations, incidents, account_transactions, audit_log	Client generates a UUIDv7 and writes locally. The push is an idempotent upsert on that id. These records CANNOT conflict, which is precisely why the records that matter most were designed this way.
Server authoritative	bookings, host_allowances, memberships, applications, firearms.status, tiers	Server holds truth. Local edits are re-applied on reconnection and may be rejected. Rejections surface to the officer — never silently discarded.

8.3 The allowance race

NAMED DEFECT TO DESIGN AGAINST

Two devices, one allowance

A member books a guest in the portal while the desk, offline, authorises a different guest against the same allowance. Both succeed locally. Both push.

Resolution: host_allowances.used_count is mutated ONLY server-side, inside a transaction, with the invitation write. Offline guest authorisation at the desk is an explicit officer override, recorded as an exception, and reconciled on sync.

If the reconciliation finds the allowance exceeded, the visit stands — the guest is already on the premises — and the overage is charged to the host. The system never resolves a data conflict by embarrassing a member.

8.4 The outbox

- A durable queue in IndexedDB. Ordered, replayed on reconnection, exponential backoff, poison-message quarantine after repeated failure with an alert to the founder.
- Depth and last-sync time are visible on the desk at all times. Silent queues are how data is lost.
- The queue survives tab close, app close, device restart and power loss.

8.5 The acceptance test

Pull the power. Do not simulate it.

On a real tablet, with no network: check in a member and a guest, sign a waiver, issue a serialised firearm and forty rounds, record two scores, then pull the power mid-session.

Restore power. Reopen. Every record must be present. Reconnect. Every record must reach the server exactly once.

This test is the definition of done for the offline layer. Browser devtools offline mode does not substitute for it and must not be accepted as evidence.

SECTION 9

Integrations

Service	Use	Requirements
Paystack	Subscriptions, guest overage, account top-up.	Webhook-driven state, never client-reported success. Idempotent on gateway_reference. Reconciliation job for webhooks that never arrive. Amounts in kobo throughout.
WhatsApp Business API	Booking confirmations, guest links, expiry warnings, conversion prompts, lapse outreach.	All outbound queued and retried. Templates versioned. Delivery status recorded. This is the primary channel — not a fallback for email.
SMS	One-time passcodes only.	A Nigerian provider with acceptable delivery rates. Test on all major local networks before launch; this is the front door to every account.
Object storage	Licence scans, member photographs, waiver signature images.	Private buckets. Signed, expiring URLs. Never public.
Targeting system	Score import.	ADAPTER ONLY, behind an interface. Not selected — see Section 14. Manual capture is the primary path and must be complete without it.
Point of sale, accounting	Bar, snack, books.	Phase 2. Manual for the first six months by explicit decision.

SECTION 10

Security and data protection

This system holds a concentration of sensitive personal data — identity documents, photographs, addresses, dates of birth and firearm licences — for guests as well as members. Nigerian data protection obligations apply and are being confirmed with counsel. Build to the requirements below in the meantime.

Requirement	Detail
Encryption	TLS in transit. Encryption at rest for database and object storage.
Document access	Licence scans readable only by the founder role. Not by range officers. Not on the desk or lane surfaces.
No shared logins	Every staff action attributable to a named person. Device-bound sessions with a short local unlock. Shared accounts destroy the attribution the custody log exists to provide.
Device revocation	A lost or stolen tablet can be revoked server-side, and its cached day pack rendered unusable on next launch.
Guest data	A guest who does not join has provided data for a single visit. A defined retention position is required from counsel; build the deletion path now.
Backup	Automated daily off-site. A restore must be performed and documented BEFORE launch. A backup never restored is a hope.
Rate limiting	On OTP request and on the guest token endpoint.
Audit	Append-only, covering every gated action and every override.

SECTION 11

Delivery plan

Sequenced so that each milestone is independently demonstrable and so that the riskiest work is proved early rather than discovered late.

Milestone	Contents	Days
M0 — Foundations	Schema, migrations, environments, CI, OTP auth, staff roles, device registration, capability service skeleton, audit log.	20–28
M1 — Offline spine	Service worker, IndexedDB, day pack, outbox, idempotent push, sync status UI. Proved with one real workflow end to end, power-pull tested.	25–35
M2 — System of record	People, memberships, tiers as permission matrix, waiver versioning and signature, qualifications, licences with expiry.	25–32
M3 — Admission	Applications, review, decisions, waitlist, roster cap enforcement, onboarding.	15–20
M4 — Hosting and booking	Availability, bookings, allowances, guest invitations, the guest link. The commercial core.	30–40
M5 — Desk	Today, status computation, check-in, host-presence rule, equipment issue, session close, end of day.	22–30
M6 — Armoury	Firearm register, custody log, derived status, ammunition lots and reconciliation, exception reporting.	25–32
M7 — Lane	Relay view, shooter-to-firearm pairing, manual score capture, incidents.	14–18
M8 — Money	Charges, Paystack, webhooks, accounts and ledger, host-pays enforcement.	18–24
M9 — Dashboard	Metrics, exceptions, expiries, overrides.	10–16
M10 — Hardening	Security review, load and period-boundary testing, restore rehearsal, staff training, seeded go-live.	18–25

Milestone	Contents	Days
	SEQUENCING NOTE M1 comes second on purpose Offline is not a phase to be added at the end. It is a property every subsequent milestone must be built into. Attempting it after M5 means rewriting M2 through M5. M1 should be demonstrated with a real workflow and a real power cut before M2 begins. If it cannot be, the schedule is wrong and it is better to know in week five than in month four.	

Total: approximately 225–315 person-days. At two developers and a designer this is a four to five month build. The stated opening is sooner. That gap is real and should be addressed by moving the date, adding people, or reducing Phase 1 scope — not by compressing M1 or M10, which are the two milestones a rushed project always sacrifices and the two that determine whether the system can be trusted.

SECTION 12

Definition of done

A milestone is complete when all of the following hold. Not when the screens exist.

- Every gated action has an automated test for both the permitted and the blocked path, and the blocked path asserts the reason string.
- Every offline-capable workflow has passed the physical power-pull test in Section 8.5.
- Every append-only table rejects update and delete at the database level, not merely in application code.
- Every write endpoint has a replay test proving one row results from two identical deliveries.
- The desk Today screen renders in under one second from cold with no network, on the actual tablet model being purchased.
- The guest link completes in under two minutes on a mid-range Android over throttled 3G.
- Seed data at realistic volume exists on staging and the milestone has been demonstrated against it.
- A range officer who has not seen the screen before can complete the workflow without instruction.

12.1 Test scenarios that must exist

Scenario	Expected
Guest arrives before host	Blocked with a clear reason. Clears automatically when the host checks in — no retry needed.
Member licence expired yesterday	Own-firearm use blocked. Club-equipment shooting still permitted. Warnings were sent at 60, 30 and 7 days.
Allowance exhausted mid-booking	Overage offered in the portal with the price shown. Never discovered at the desk.
Session close with a firearm unreturned	Blocked. Lists the specific firearm. Override requires a reason and appears on the dashboard.
Two devices authorise against one allowance offline	Both visits stand. Overage charged to the host. Exception raised. No member embarrassed.
Roster at cap, application admitted	Blocked, or admitted only with explicit founder override recorded.
Duplicate webhook from Paystack	One payment row. Idempotent on gateway_reference.
Guest shoots a member-owned firearm	Blocked at pairing. If overridden, appears as a same-day dashboard exception.

SECTION 13

Out of scope for Phase 1

Named explicitly so nobody builds them speculatively and nobody assumes they were forgotten.

- Leagues engine, ladder, standings, fixtures — Phase 2. Score capture in Phase 1 exists to populate them.
- Electronic score import — adapter interface only until the targeting system is selected.
- Bar and snack point of sale; accounting integration; merchandise — manual for six months by explicit decision.
- Range officer rostering — WhatsApp and a spreadsheet.
- Framed-target workflow, live tournament display, member-to-member messaging, native apps, multi-currency, multi-site.
- On-premises firearm storage workflow — tables built, workflow behind a flag pending the decision in Section 14.

SECTION 14

Blocking and non-blocking open items

Item	Blocks	Needed by
Regulatory and licensing obligation	Schema freeze on custody and ammunition; retention policy; guest data position.	Before M6. Counsel engaged now.
Lanes per discipline, session length, opening hours	Availability computation, roster cap.	Before M4.
Roster cap and capacity model	Admission enforcement, waitlist.	Before M3.
Tier names, fees, guest allowances	Seed data only — the matrix can be built without them.	Before M10.
Guest overage price	Charging logic. Configurable, so not structurally blocking.	Before M8.
On-premises storage decision	One workflow behind a flag.	Before M6.
Member-supplied ammunition	Reconciliation logic.	Before M6.
Targeting system selection	Nothing. Deliberately de-gated — manual capture is primary.	Phase 2.

Three things to hold on to

For the team

1. Offline is not a feature. It is a property of everything, proved with a real power cut on a real tablet, before the rest is built on top of it.
2. Append-only is not a preference. Custody events, waivers and scores are never edited and never deleted — enforced at the database, not in application code. These are the records the club's licence and its entire product both rest on.
3. Fifteen seconds at the desk, twenty at the lane, two minutes on the guest link. Every screen is measured against a clock, because the club has no administrator and never will. A workflow that exceeds its budget will be abandoned, and staff reverting to paper is how systems like this actually fail.